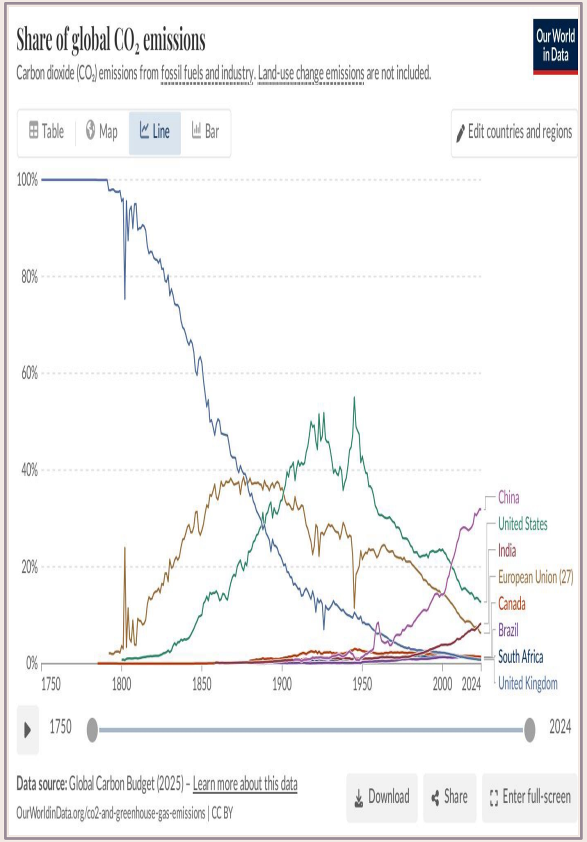
# Global $CO_2$ Emissions: Who Emits, How Much, and What's Changing

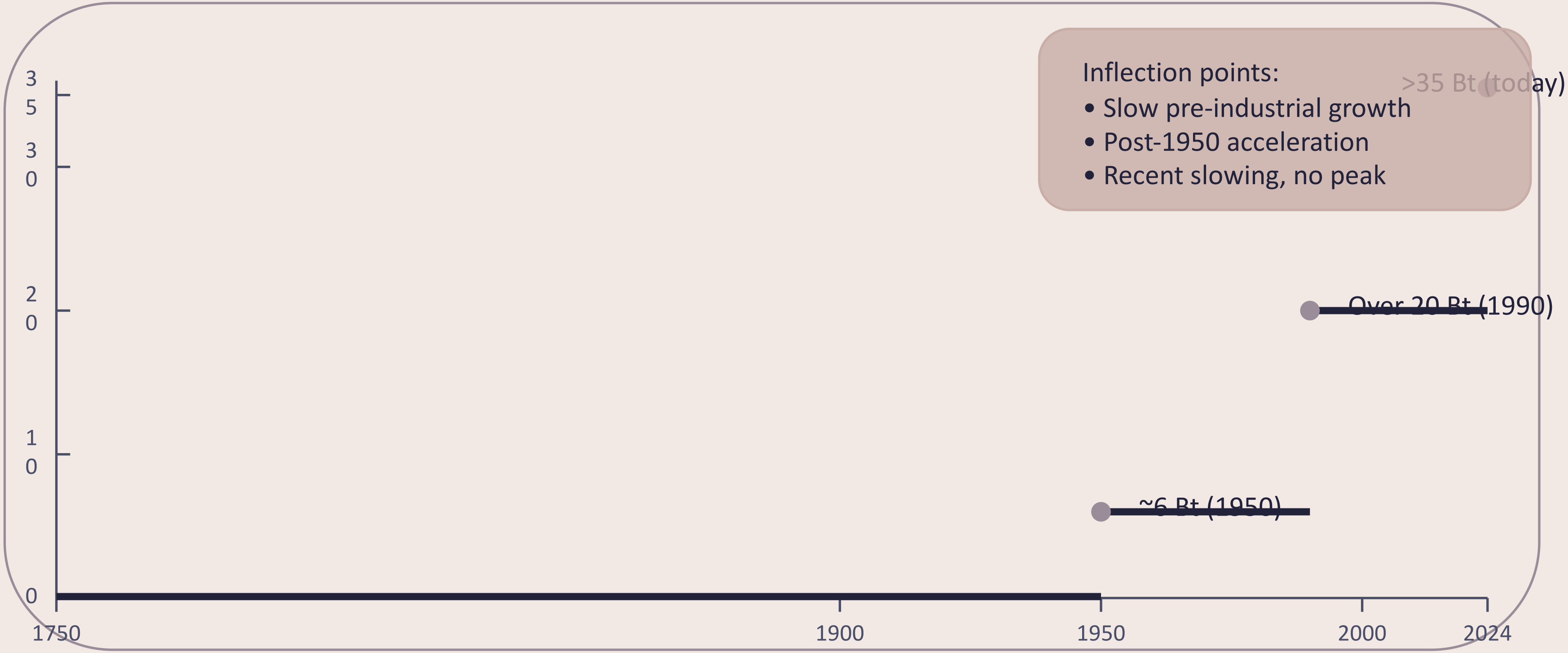
Fossil fuels & industry • 1750–2024

Data: Global Carbon Budget 2025 (via Our World in Data)

For policy analysts, climate negotiators, and sustainability researchers



# Global $CO_2$ Emissions Grew from 6 Bt in 1950 to Over 35 Bt Today




Source: Global Carbon Budget 2025 via Our World in Data (fossil fuels & industry).

# Europe and the US Fell from 90% of Emissions in 1900 to Under One-Third Today

Stacked area chart (share of global $CO_2$ over time)

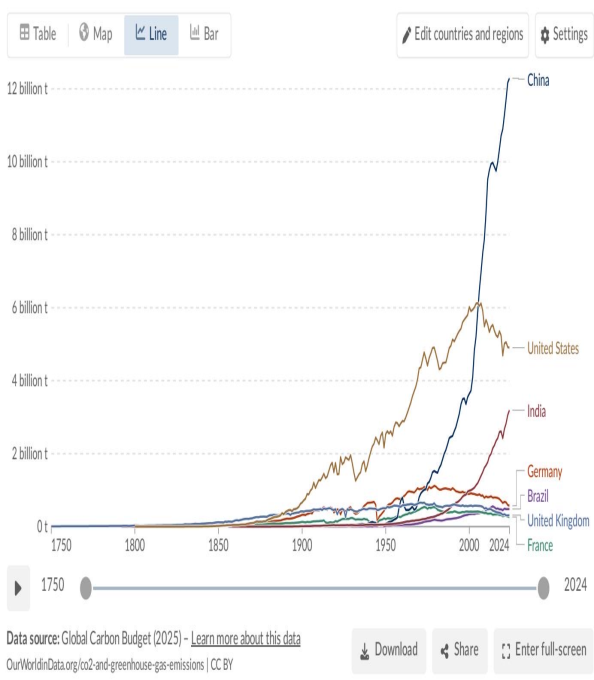
Asia (led by China)

Europe

United States

Rest of world

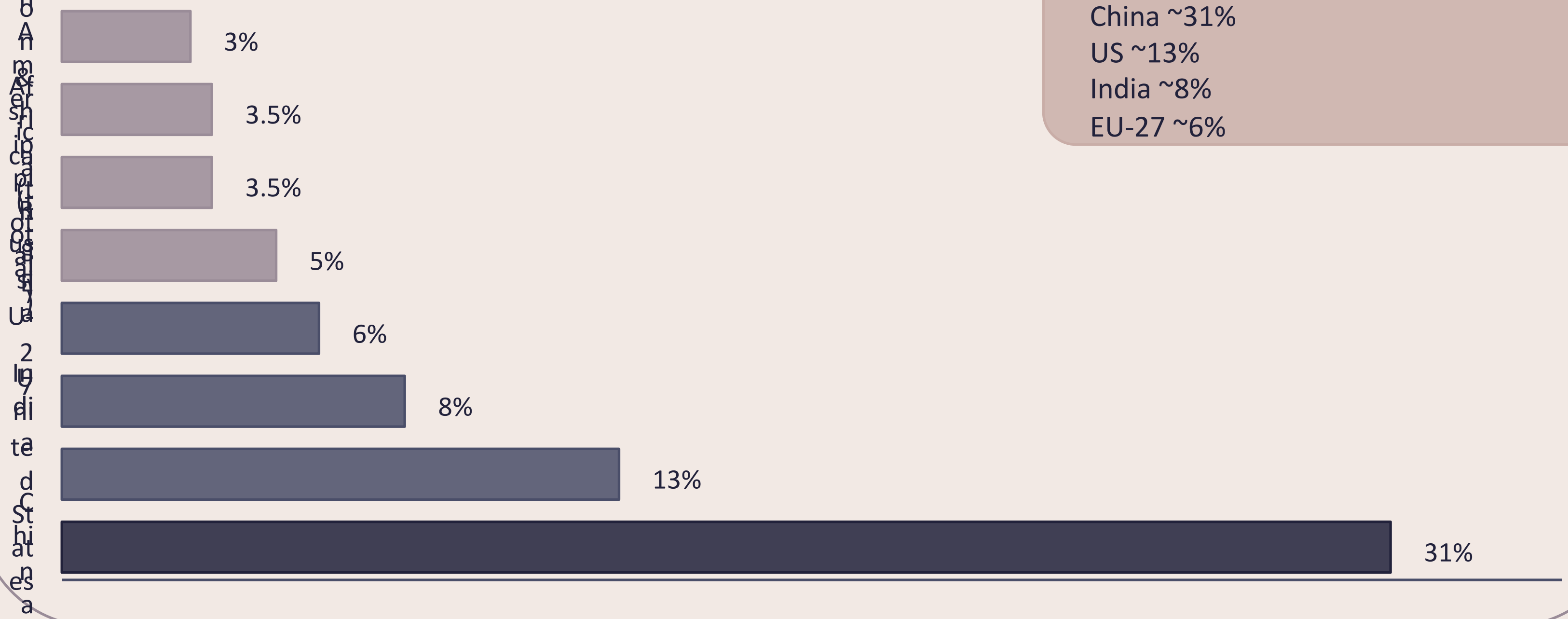
1900: Europe + US > 90% • Today: < one-third • Asia ≈ half




Visual reference: OWID Figure (share over time)

Source: Global Carbon Budget 2025 via Our World in Data.

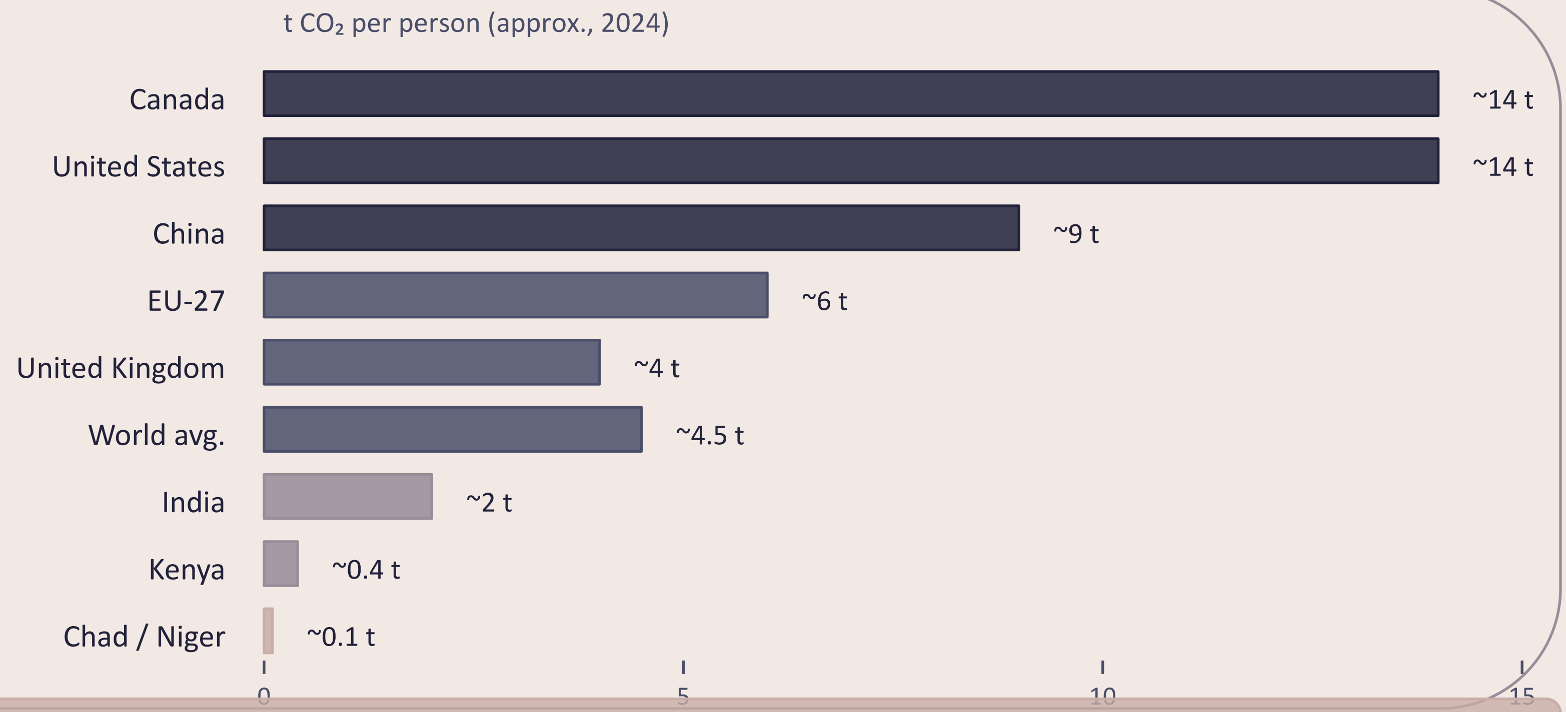
# China Now Emits More Than One-Quarter of Global $CO_2$




2024 shares (approx.):
China ~31%
US ~13%
India ~8%
EU-27 ~6%

Source: Global Carbon Budget 2025 via Our World in Data (approx. shares, 2024).

# Per Capita Emissions Range from 0.1 t in Chad to ~14 t in the US and Canada

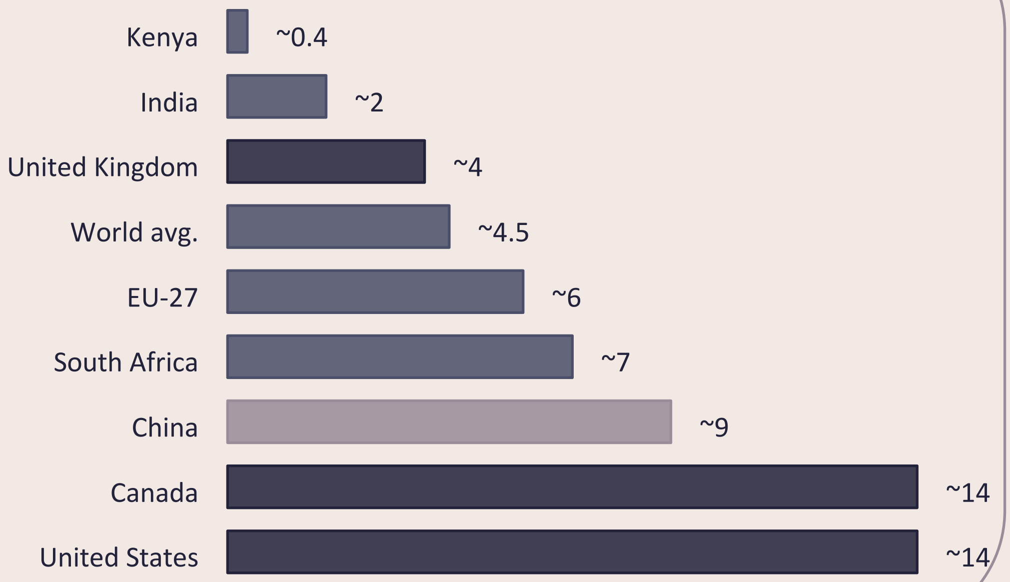



~150× gap: ~0.1 t/person (lowest) vs ~14–15 t/person (US/Canada/Australia)

Source: Global Carbon Budget 2025 via Our World in Data (approx. per capita, 2024).

# Per Capita Emissions Vary Widely: ~14 t in the US vs. ~9 t in China and ~2 t in India

Horizontal bar chart (approx., 2024)



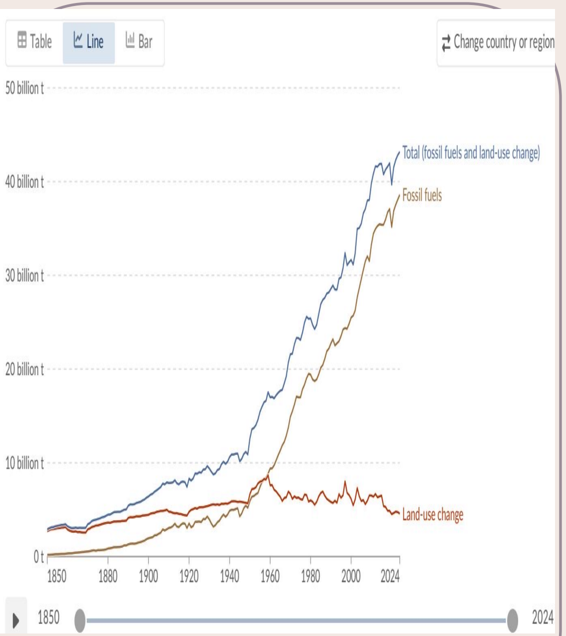
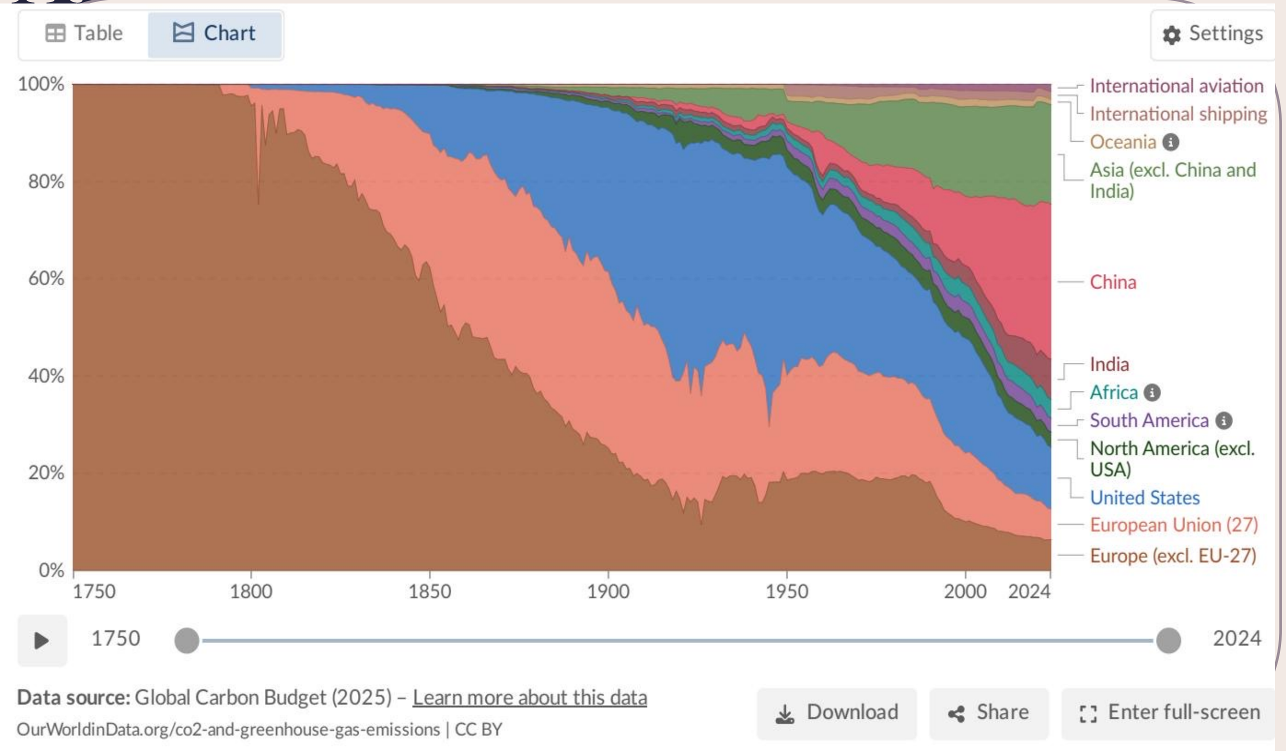




Figure reference: per-capita trajectories over time

Largest total emitter ≠ highest per-capita: China ~9 t/person is below US/Canada ~14 t/person.

Source: Global Carbon Budget 2025 via Our World in Data (per capita, approx. 2024).

# 2024 Snapshot: Many Western Nations Cut Emissions While Parts of Asia and Africa Grew




## Interpretation (2024)

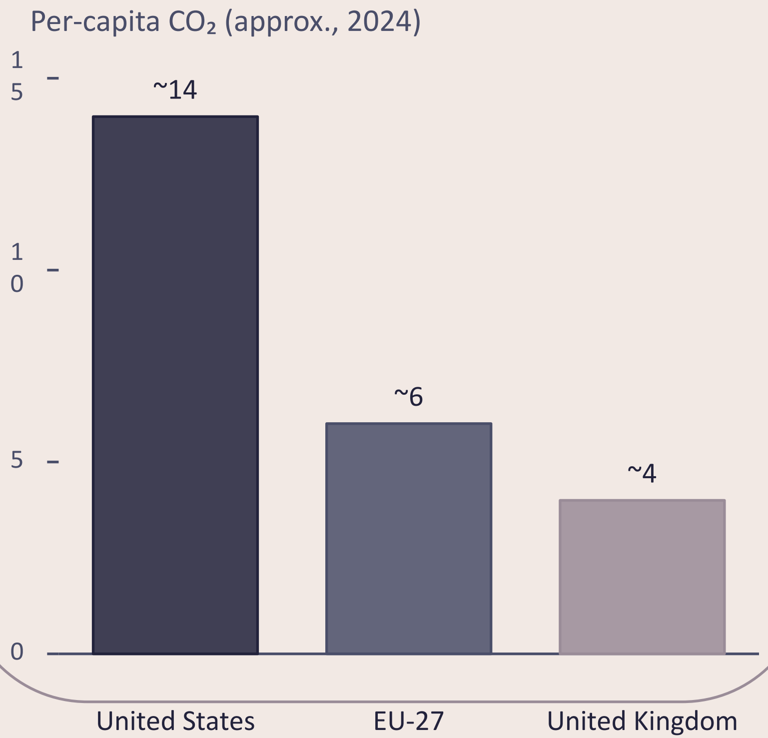
- US, much of the EU, and the UK: declines (blue)
- Russia and parts of Southeast Asia: increases (orange/red)
- Several African nations: increases
- Some Central Asia countries: increases >10%

Decline

Increase

Source: Global Carbon Budget 2025 via Our World in Data (annual % change in $CO_2$ emissions, 2024).

# Energy Mix Explains Why France and the UK Emit Far Less Per Capita Than Germany or the US




## Why differences persist among wealthy countries

Per-capita emissions can diverge even at similar income levels.

France, the UK, and Portugal have much lower per-capita emissions than Germany, the Netherlands, or the US, largely because electricity relies more on nuclear and renewables rather than fossil fuels.

Policy lever: power-sector decarbonization can cut per-capita emissions without sacrificing living standards.

Source: Global Carbon Budget 2025 via Our World in Data (per capita figures; energy-mix insight from OWID synthesis).

# Key Takeaways: Responsibility Is Shared but Unequal

| | |
|---|---|
| **Scale** | Global $CO_2$ emissions from fossil fuels exceed 35 Bt/yr and have not yet peaked (growth has slowed). |
| **Who emits now** | China is the largest annual emitter (~31% of global $CO_2$), but its per-capita emissions (~9 t) remain below the US (~14 t). |
| **Historical shift** | Europe + the US dominated emissions historically (>90% in 1900) but now account for less than one-third. |
| **Inequality** | Per-capita gaps of ~150× persist between the lowest emitters (~0.1 t/person) and high-emitting large countries (~14–15 t/person). |
| **Policy lever** | Energy mix choices can decouple prosperity from emissions (e.g., UK far below US |

Source: Global Carbon Budget 2025 via Our World in Data